

Drop by Drop: Flow of Hope

Game Design Document — Full Game Flow & Systems Reference

Genre: Four-lane endless runner with resource survival and village restoration

Platform: Unity (PC)

Document purpose: Give the team a single reference for game flow, mechanics, level design, and technical architecture.

Generated from the live codebase — reflects current implementation.

1. High Concept

The player guides Zuri through three runner levels in a drought-stricken village. Each level teaches and tests a different survival skill while physically restoring the community's water infrastructure. Progress is measured as **Village Restoration %** displayed on the HUD.

Milestone	Village %	Narrative meaning
Level 1 complete	0 → 35%	Build the village well — collect materials, fill the bucket
Level 2 complete	35 → 65%	Create the water bowl — survive mudlands hazards
Level 3 complete	65 → 100%	Repair three water tanks before time runs out

Core fantasy: Every drop of water collected, every material gathered, and every pipe repaired visibly restores hope to the village.

2. Scene Flow & Progression

Scenes are defined in **SceneCatalog.cs**. The player journey:

StartScreen → **MainGame (Level 1)** → **Level2** → **Level3** → Victory / End

Scene	Unity name	Role
Start Screen	StartScreen	Main menu — no gameplay bootstrap
Level 1	MainGame	Desert tutorial + water survival
Level 2	Level2	Mudlands hazards + warthogs
Level 3	Level3	Timed pipe-repair sprint
Info	StarterInfor	Optional tutorial/info screen

Level transitions

- Victory panel appears when win conditions are met at the finish gate.
- Player clicks **Next Level** → `RunStateManager.GoToNextScene()` loads the next scene.
- Pause menu offers Restart, Main Menu, and dev shortcuts to Level 2/3.
- Death pauses the game (`timeScale = 0`) and shows the lose panel with reason text.

3. Core Player Mechanics

3.1 Movement (PlayerController)

Input	Action
A / ←	Move one lane left
D / →	Move one lane right
Space	Jump (force 15, 0.18s lock)

Four lanes (indices 0–3). Default X positions: –6, –2, 2, 6 (re-centered on ground at runtime). Player starts in lane 2. Forward speed is set per level by pacing scripts (~18–24 m/s).

3.2 Resources (PlayerResources)

Resource	Max	L1 start	L2 start	L3 start
Health	100	100	100	100
Player Water (body)	100	100	100	100
Bucket Water	100	15	0	25
Materials	100	0	0	0

Player Water = hydration for the character. **Bucket Water** = water carried for the village. Both matter for survival and village progress.

3.3 Village Progress (VillageProgressService)

- Level 1: $((\text{materialNorm} + \text{bucketNorm}) / 2) \times 35$ — capped at 35%
- Level 2: $\text{Lerp}(35, 65, (\text{materialNorm} + \text{bucketNorm}) / 2)$
- Level 3: $65 + \text{average}(\text{tank1}, \text{tank2}, \text{tank3}) \times 0.35$

L1 bucketNorm uses growth from starting 15 → 100. Materials and bucket both contribute equally.

3.4 Win & Lose (RunStateManager + HUDControls)

Condition	Levels	Result
Health ≤ 0	All	Death
Player water = 0 for 18s	L1 primary	Death (low-water grace)
Time limit expires	L3	Death
Finish gate without full materials + 1 bucket	L1 + L2	Death with reason
Finish gate without all tanks at 100%	L3	Death with reason
All win checks pass at gate	All	Victory panel → next level

4. Level 1 — Desert / Dry Bushlands (MainGame)

Purpose: Teach four-lane movement, jumping, water survival, and resource collection. Village progress caps at 35%.

Run length: ~1,015 m | **Target time:** ~3:30 | **Layout:** Level1LayoutDirector

4.1 Teaching Phases

Phase	Progress	Player learns
Learn	0–25%	Lanes, jump, cactus water, basic obstacles — no snakes/heat yet
Practice	25–50%	Heat waves, snakes, water management
Combine	50–75%	Mix hazards + rolling logs + recovery
Challenge	75–100%	Higher frequency, always a safe lane + recovery nearby

4.2 Key Systems

System	When	Effect
Heat wave	25% progress	3–5s bursts, –20 player water/sec, 7s cooldown
Snakes	From ~28%	Lane warning → charge; –2 player water on hit
Cactus (water source)	Throughout	Pass-through: +18 body, +10 bucket / 1s
Water springs	14 placed pools	+20 body, +12 bucket / 0.9s while standing
Rolling logs	From ~52%	Span 2 lanes — jump to avoid; –5 health on hit
Obstacles (rock/sand)	Throughout	Bucket damage; sand/rock slow + material loss
Black pits	Mid/late	–5 health + slow

4.3 Win Condition

- Reach finish gate (Ender1) with Health > 0, Player Water > 0, Materials = 100, Bucket = 100.
- Victory message: "LEVEL 1 COMPLETE" → option to start Level 2.

4.4 Tutorial & Feedback

- Level1TutorialUI — tips until 25%: A/D lanes, materials, cactus, jump, heat, snakes, logs.
- Level1FeedbackUI — toasts for water collect, damage, heat, snakes, low water.
- Level1LowWaterMonitor — warns at ≤35% and ≤15% player water.

5. Level 2 — Mudlands (Level2)

Purpose: Apply L1 skills under mud, poison, tracking enemies, and environmental hazards. Village progress 35% → 65%.

Run length: ~1,414 m | **Target time:** ~3:00 | **Layout:** Level2LayoutDirector

5.1 Layout Bands

Band	Progress	Content
Tutorial	0–15%	Mud, rocks, cactus, water pickups
Warthogs intro	15–30%	First warthog crossings (speed 36)
Logs + poison	30–45%	Rolling logs, poison plants, warthogs (38)
Mid-run	45–60%	Mud monsters, denser hazard mix
Mixed pressure	60–75%	Warthogs 40–42, bubble shields
Finish stretch	75–100%	Warthogs 43–45, extra recovery pickups

5.2 Hazards

Hazard	Avoidance	Penalty
Warthog	Jump (homes onto your lane)	–10 health
Mud puddle	Jump or lane change	Slow ×0.65 for 2s
Mud monster	Dodge mud balls; bubble shield blocks	–5 health, slow ×0.45
Poison plant	Avoid 2-lane gas zone	Health + water drain over time
Rolling log	Jump	–5 health
Cactus	Avoid collision	–10 health, –5 player water
Rock	Jump or dodge	–10 health

5.3 Pickups

Pickup	Values
Water droplet	+15 player water, +15 bucket
Water pool	+10 player, +25 bucket
Baobab	+20 player water
Material	+15 (Pipe / Nails / Hammer cycle)
Health fruit	12 / 15 / 18 / 20 (rotating)
Bubble shield	9s — blocks mud balls and poison

5.4 Win Condition

- Same as Level 1: 100/100 materials, 100/100 bucket, health and player water > 0 at Ender2.

6. Level 3 — Pipe Repair Sprint (Level3)

Purpose: Timed sprint while spending materials to repair three water tanks. Village progress 65% → 100%.

Run length: ~7,182 m | **Hard time limit:** 180 seconds (3:00)

6.1 Primary Objective — Tank Repair

Tank	Repairs needed	Pipe chances	Material/repair	Progress/repair
Tank 1	5	8	4	+20%
Tank 2	10	15	5	+10%
Tank 3	2	5	6	+50%

Tank HUD positions along run: ~40%, ~76%, ~90%. Player presses repair key at pipe spawn points while carrying enough materials.

6.2 Wave Schedule (Level3WaveDirector)

Progress	Wave mode
0–20%	Intro hazards
20–50%	Snake wave
50–75%	Warthog wave
75–90%	Snake wave
90–95%	Combined hazards
95–100%	CombinedHard + boss sequence

6.3 Hazards (selected)

- Lightning: 2s warning → –5 health, –5% materials
- Acid rain: 2s warning → –10 health, –5% materials
- Snake: –3 health | Warthog: –5 health | Rolling log: –6 health, –5 materials
- Mud: –5 materials | Tree: –10 materials, –5% bucket

6.4 Win / Lose

- Win: All three tanks reach 100% (can finish before or at end gate).
- Lose: Time expires, or finish reached with incomplete tanks.
- Level3FeedbackUI shows tank percentages, time warnings, and wave transitions.

7. Technical Architecture (for developers)

7.1 Bootstrap Pipeline (SceneBootstrapper)

On every gameplay scene (not StartScreen/StarterInfor), execution order –100:

- RunStateManager — pause/death/victory state
- PlayerController — disables legacy PlayerMovement scripts
- RunnerPlayerSetup — lane alignment, forward speed, jump feel
- HUDControls + PlayerResources + VillageProgressService
- GroundSpawner — streaming ground tiles
- Level-specific LayoutDirector + FeedbackUI (+ L1 TutorialUI, LowWaterMonitor)

7.2 Layout Directors

Each level builds all gameplay objects at runtime — no broken legacy Spawner/BucketSpawner. Scene-placed pickups are disabled; fresh layout is generated from progress-percent beats.

Director	Scene	Spawns
Level1LayoutDirector	MainGame	Materials, cactus, springs, snakes, logs, obstacles, finish gate
Level2LayoutDirector	Level2	Mud, warthogs, poison, monsters, logs, pickups, finish gate
Level3LayoutDirector	Level3	Pipes, hazards, waves, pickups, tank markers

7.3 Key Scripts Reference

Script	Responsibility
PlayerController	Lane movement, jump, speed modifiers
PlayerResources	Health, water, bucket, materials — single source of truth
HUDControls	UI display, win/lose checks, resource API for pickups
RunStateManager	Playing/Paused/Dead/Victory states, scene transitions
VillageProgressService	Village % formulas per level
SceneCatalog	Canonical scene names and helpers
Level1/2/3FinishGate	End trigger → LevelProgress()

8. Design Pillars & Player Experience Goals

Pillar	How it shows up
Teach, then test	L1 phases introduce one mechanic at a time before combining
Water is life	Body water + bucket water both matter; heat and hazards drain hydration
Visible restoration	Village % rises as player collects and repairs infrastructure
Fair difficulty	Every hazard beat has reaction time, open lanes, or recovery nearby
Four-lane readability	All objects placed in lanes 0–3; patterns use lane choice as gameplay
No random unfairness	Layout directors place intentional beats — not frame-by-frame spawning

9. Quick Playtest Checklist

- L1: Player learns A/D and Space; cactus refills water; heat manageable; village caps at 35%; Level 2 loads.
- L2: Warthogs require jump; mud/poison readable; materials + bucket reach 100 at gate; Level 3 loads.
- L3: All three tanks repairable within 3 minutes; time warnings appear; village reaches 100%.
- All levels: No console errors; pause/restart/menu work; death shows clear reason.